

TO: ME 2171 Instrumentation Lab Instructional Staff
FROM: Mateo Jaime Tecum Jacinto, Section 07- Wednesday
SUBJECT: Lab 4, Strain Gages For Strain & Stress Measurement

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1.0 Statement of Purpose

The objectives for this lab included following the procedures of the strain gauge setup, which includes degreasing, abrasion, burnishing, conditioning, and neutralizing, and gage preparation, gage transfer, applying catalyst and then supergluing and removing the tape.

Additionally, after setup, connecting the strain gauge to the signal conditioner (blue box with a quarter bridge setup) to complete the load cell to obtain readings/outputs of strain from adding masses at the end of the cantilever beam and connecting it to our DAQ board and VI, so we could get readings of voltages, strain, and weight from the programmed VI.

2.1 Background

The strain gauge is an analog sensor that can record deformation of an object (strain) as a change in its own electrical resistance. The change is calculated from the given equation: $\frac{\Delta R}{\epsilon} = GF$, and the gauge factor is set to 2.21. The strain on the cantilever will cause the strain gauge to also deform; this deformation in a conductor causes the resistance to change as it changes shape. Knowing $GF = 2.21$, the change in resistance can allow for a calculated ϵ value.

A strain gauge can be used to determine a load cell by first adhering a strain gauge to the beam and then zero out the strain value on the blue box setup (which with the strain gauge acts as the load cell). Then by adding the force as a weight on the beam, our strain gauge records the change in its electrical resistance, which is then recorded as a digital voltage due to the blue box setup using a Wheatstone bridge circuit to make the small change in resistance a distinct voltage change, which is displayed on the box.

The quarter bridge setup uses the strain gauge as one leg of the Wheatstone bridge. Then the other legs are set to the internal resistors on the blue box, that way our only change will be the resistance in the strain gauge. By having the set up auto balanced, the voltage is zero when there is no weight applied. This accounts for voltage offset that occurs from the strain gauge itself from any initial resistance caused during setup. So, once any weight or other causes of strain occur, a nonzero voltage value will be shown. The nonzero voltage is then amplified through the A/D converter by the signal conditioner. Using the VI and the A/D converter, we can use this output voltage and convert it to a digital value (a count), which can be programmed to obtain a strain value on the VI front panel, which displays the strain in micro units, and force into pound force.

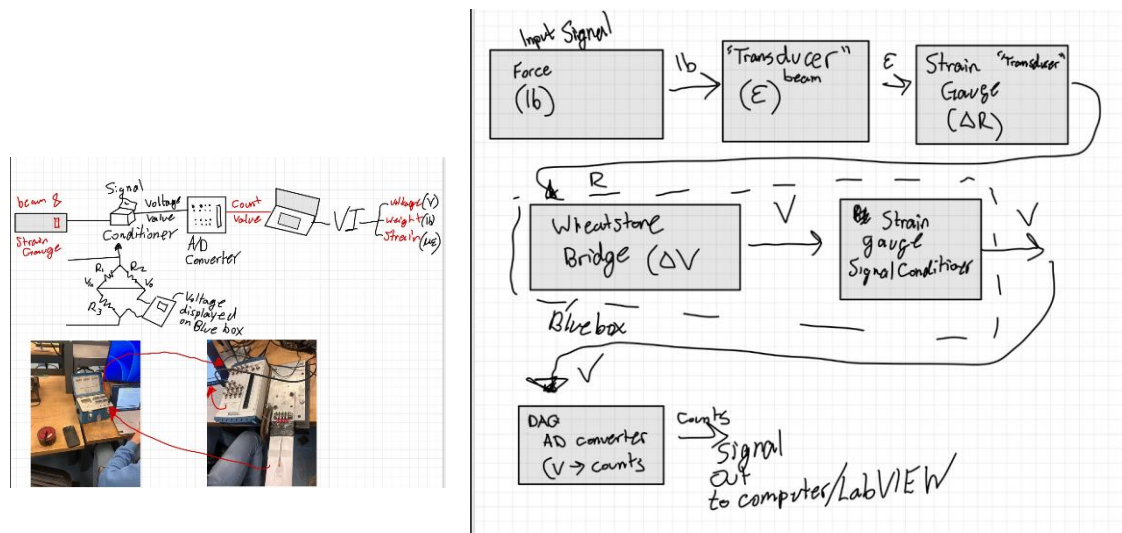


Figure 1 diagram of set-up with pictures and flow chart.

3.0 Discussion

3.1 Equipment

- LabVIEW (computer that runs LabVIEW)
- NI-USB-6221 (DAQ) board
- Paper towels
- Cotton Swabs
- Neutralizer – MCA-5
- Conditioner – MCS-3
- Degreasing agent/spray – CSM-1
- Strain gauge
- Tape
- Superglue (did not use M-Bond catalyst)
- Aluminum Cantilever beam

- Wires (on cantilever beam)
- Blue box – p3 strain indicator and recorder
- BNC cable
- Ruler
- 220 grit Sandpaper
- 1, 2.5, and 5lb plates
- Plate holder (metal hanger)

3.2 Calibration procedure

The strain gauge force transducer, which was the beam in this lab, was calibrated on the blue box after having the set up with the plate holder on the end of the beam and the strain gauge already on. Once everything was set, going to the settings of the blue box and ensuring both the setup and the output BNC wire was connected, our team used the auto balance function to zero out the strain and voltage that was displayed on the blue box display screen. This made it so the beam with everything on is the reference for our other strained values. This is necessary to do as it will tare out the strain caused by the plate holder on the end of the cantilever beam, so that when we get our strain values and weight, it will not include the strain, nor the weight caused by the plate holder. Then the calibration procedure included applying 1 lb weights and recording the microstrain readings from the blue box display. This was done starting from 0 up to 10 lbs. Channel 3 was used on the blue box in our setup.

The beam dimensions were as follows:

X(distance from load to strain gauge): 52.3 cm

B(width of the beam): 6.7 cm

T(thickness of the beam): 0.96 cm

Equation 1. Theoretical Strain Equation: $\varepsilon = \frac{6FX}{Ebt^2}$

In this strain equation

ε represents the strain measured in m/m, which will be converted to microstrain

F representing the applied force in Newtons (N)

X representing the distance from the load point (the bolt) to the strain gauge location in meter(m)

E represents the modulus of Elasticity of Aluminum in Pascals (Pa), which is 6.9×10^{10} Pa (or 10.0×10^6 psi)

b representing the width of the aluminum beam in meters(m)

t representing the thickness of the aluminum beam in meters(m)

Theoretical Strain calculations:

$$\varepsilon = \frac{6(4.448N)(0.523m)}{(6.9 \times 10^{10}Pa)(0.067m)(0.0096m)^2}$$

$$\varepsilon = 3.28 \times \frac{10^{-5}m}{m} = 32.8\mu\varepsilon$$

$$\varepsilon = \frac{6(8.9N)(0.523m)}{(6.9 \times 10^{10}Pa)(0.067m)(0.0096m)^2}$$

$$\varepsilon = 65.6\mu\varepsilon$$

$$\varepsilon = \frac{6(22.24N)(0.523m)}{(6.9 \times 10^{10}Pa)(0.067m)(0.0096m)^2}$$

$$\varepsilon = 163.89\mu\varepsilon$$

Data sheet:

Applied Weight(lbs)	Force(N)	Theoretical Strain(micrstrain)	Experimental(microstrain)
1	4.45	32.77957746	31
2	8.9	65.55915493	62
3	13.35	98.33873239	92
4	17.8	131.1183099	122
5	22.25	163.8978873	147
6	26.7	196.6774648	184
7	31.15	229.4570423	214
8	35.6	262.2366197	245
9	40.05	295.0161972	275
10	44.5	327.7957746	305

Table 1. Raw Data Theoretical and Experimental Strain

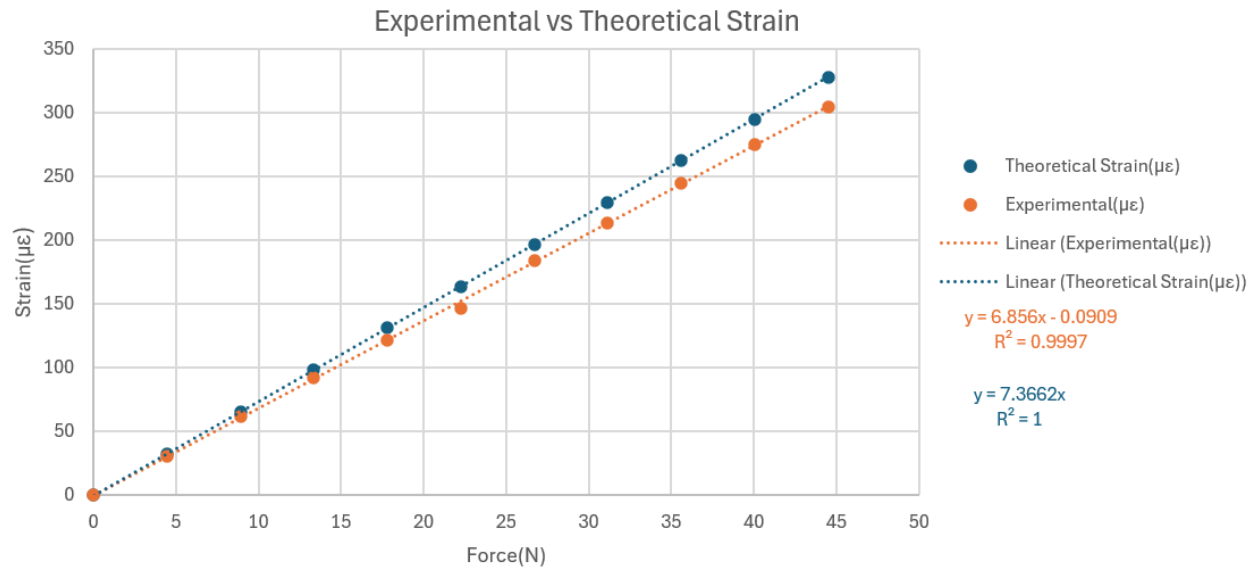


Figure 2. Graph of measure strain and calculated strain (load force(N) versus strain (microstrain) with two series, experimental and theoretical strain

The graph serves as enough evidence to claim that the theoretical and the measured strain are in agreement as the data points following from the calculations and data measured are both showing a linear relationship of the force and strain and the data points of each are pretty similar in value, which would make sense as our measurements most likely have some uncertainty values that cause this variability between the data sets. The reasoning for why these are different is primarily due to the tared weight giving us the incorrect offset voltage and our strain gauge not being perfectly parallel. The strain gauge not being perfectly aligned with the strain axis of the cantilever beam would cause the change in resistance to be a different value than what it should be causing measurements to be off. The offset voltage being off by a small amount would also cause a small voltage to result in different values of strain than what was calculated. Another reason for this invariability is that the weight of each plate is not exact. Some of the plates used showed worn usage which could cause them to no longer be exactly 1 or 5lb forces being caused on the beam. Additionally, our measurements of the beam are also not exact, specifically the measurement from the end of the cantilever beam to the strain gauge. This measurement is important, and our measured value was only taken to be ~50 cm, which does not show the precise distance.

3.3 LabVIEW vi for Weight Measurement

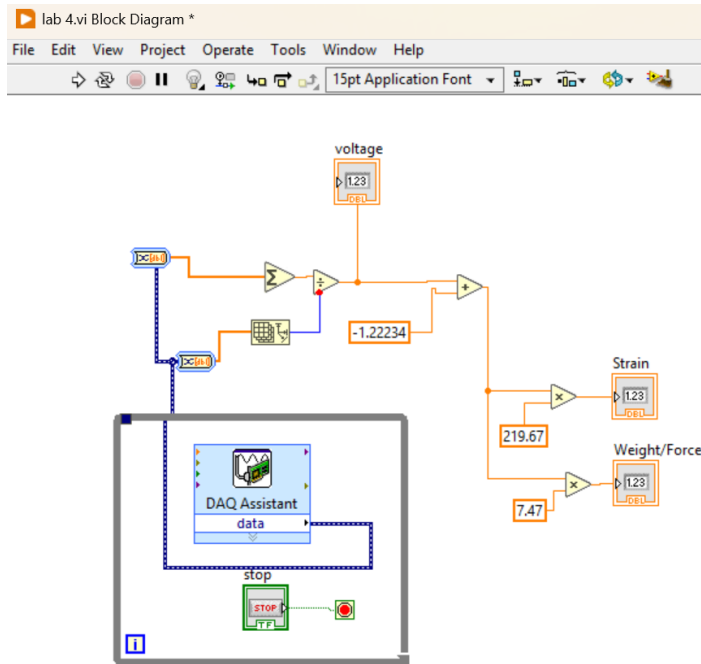


Figure 3. Block diagram showing data stream from DAQ (count value) to output voltage, strain, & weight

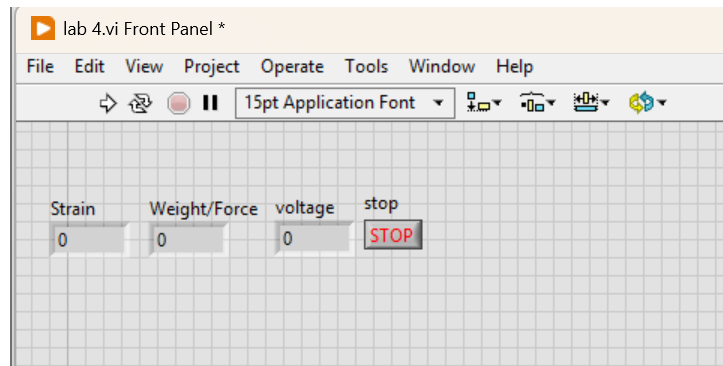


Figure 4. Front Panel showing strain, weight, voltage & stop

The vi works by starting off with the DAQ assistant data stream providing a raw voltage signal from the A/D converter (which originated from the strain gauge conditioner). The DAQ is setup so that the input range of the voltage is from 0 to 2.5V. The while loop works to gather lots of voltage values, which are then pass through a tunnel, which is configured to only output the last value. This value is passed through a simple function that averages out the voltage, so that any noise is minimal and does not affect the voltage value by much. The voltage is then displayed on the front panel through a numeric indicator. Then the voltage is converted to strain by adding the offset voltage and then multiplying by a conversion rate that was determined by $(\text{Strain} - 0) / (V - V_{\text{offset}})$, which was $219.67 \mu\epsilon/V$. This value of strain is then displayed on the front panel as well, and the data is then converted from voltage(from a previous point in the data stream to weight, which is done by multiplying by a conversion factor (which is a value

of 7.472) from volts to pounds. The force is then displayed in units of pound force. Although not done for us, a conversion factor of 26 could be used to convert from strain to pounds.

Conversion Factor Calculations:

$$\frac{147\mu\epsilon}{(1.89153V - 1.22234V)} = 219.67\mu\epsilon/V$$

$$Strain(\mu\epsilon) = (xV - 1.22234V) * 219.67$$

Equation 5. Voltage to Strain Conversion Factor Calculation

$$\frac{5lb - 0lb}{(1.89153V - 1.22234V)} = 7.472lb/V$$

$$Weight(lbs) = (xV - 1.22234V) * 7.472lb/V$$

Equation 2. Voltage to Strain Conversion Factor Calculation

Low limit	0V
High Limit	2.5V
Channel	AI0

Table 2. DAQ board settings

The resolution of the readout system in units of microstrain was set to 16-bits (as that is what the DAQ board has). Therefor the amount of counts is $2^{16} = 65526$ counts. The voltage range was $2.5V - 0V = \mathbf{2.5V}$. And the Voltage per count was calculated as $\frac{2.5V}{65536\ counts} = 3.81 \times 10^{-5}V/count$. The strain per volt was given above as $219.67\ \mu\epsilon/V$. to determine the resolution we can multiply $V/count$ and $\mu\epsilon/V$, which results in $\mathbf{8.37 \times 10^{-3} \mu\epsilon/count}$.

For finding the unknown mass measurement, we weighed Will's calculator and determined it was around 0.284lb.

Measured voltage for unknown mass(Will's calculator):	1.26V
Calculated Strain:	8.4 $\mu\epsilon$
Calculated Weight:	.284lb
Calculated mass:	.008826 slugs

Table 3. unknown mass measurements and calculations

4.0 Conclusions

The strain gauge setup was a successful application to obtain the force applied to the aluminum cantilever beam and also works well for as the force transducer. The experimental calibration resulted in a sensitivity of $8.37 \times 10^{-3} \mu\epsilon/\text{count}$. The LabVIEW vi was able to obtain voltage signals from the A/D converter and displayed the voltage, microstrain, and weight with some deviation.